Properties of Eigen vector

(1) If x is an E.vector for A so (kx) also and vice-versa.

(2) For different eigen values corresponding eigen vectors are LI.

(3) If E.values repeats then corresponding eigen vector may be LI or may be LD.

(4) Algebric Multiplicity (AM) = No. of times of a particular eigen values repeats is called its AM.

For $A_{6\times6}$ let $\lambda = 5, 5, 2, 2, 2, 7$

AM of $5 = 2$
AM of $2 = 3$
AM of $7 = 1$

→ Geometric multiplicity (GM) = No. of linearly independent eigen vector for a particular E.value is called its GM.

→ Eigen vectors corresponding to different eigen values of symmetric matrix are orthogonal.

→ Shortcut method of finding Eigen vectors (If options given) – λ should be unique in all comparison and there should not be any absurd result.

→ Shortcut method of finding the counting of linearly independent eigen vectors for repeated eigen value of λ :–

No. of LI eigen vectors for λ = order (n) – $\rho(A - \lambda I)$